Stress Testing a Microsoft–Apple Portfolio

Data and EWMA. I use Yahoo Finance daily closing prices, adjusted for stock splits but not dividends. The 1,259 log returns are dated September 1, 2011 through September 1, 2016; the August 31, 2011 close supplies the first return. The given market caps imply weights $(0.438303, 0.561697)$, so the \$1,000,000 portfolio has fixed initial dollar positions $\theta_t = (438{,}303.32, 561{,}696.68)^\top$, ordered as (MSFT, AAPL). Starting from zero, I update the mean and covariance with decay 0.97, using the previous mean in each covariance residual. The final estimates are

$$\mu = \begin{pmatrix} 0.00125025 \\ 0.00106069 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 0.000129897 & 0.000033007 \\ 0.000033007 & 0.000151852 \end{pmatrix}, \qquad \rho = 0.235013.$$

Initial VaR and capital. For the normally distributed one-day linearized loss,

$$\mathrm{VaR}_{0.95,1} = -\theta_t^\top \mu + 1.644854\sqrt{\theta_t^\top \Sigma \theta_t} = \$14{,}383.87.$$

Applying the requested square-root-of-time rule gives

$$\mathrm{VaR}^{\mathrm{scale}}_{0.95,10} = \sqrt{10}\,\mathrm{VaR}_{0.95,1} = \boxed{\$45{,}485.80}, \qquad C = 3\,\mathrm{VaR}^{\mathrm{scale}}_{0.95,10} = \boxed{\$136{,}457.39}.$$

Shock and simulation. Apple's five-sigma log-return shock is $x^{(2)} = -0.0605534$. Conditional on this value, Microsoft's mean and standard deviation are $\widetilde{\mu}^{(1)} = -0.0121422$ and $\widetilde{\sigma}^{(1)} = 0.0110780$. Its conditional five-sigma shock is

$$x^{(1)} = \widetilde{\mu}^{(1)} - 5\widetilde{\sigma}^{(1)} = -0.0675324.$$

Both first-day returns are fixed. After updating EWMA with this shock, I simulate days 2–10 from the current bivariate normal distribution, updating the mean and covariance separately along each path. For each of 50,000 paths (seed 731), I compute

$$\ell_m = -\theta_t^\top \sum_{k=1}^{10} X^m_{t+k\Delta}.$$

The empirical 95% VaR is the 47,500th sorted loss, without interpolation.

	Requested result		Estimate
(a)	Average stressed ten-day loss		\$70,838.04
(b)	Stressed ten-day 95% VaR		\$151,692.60
(c)	Loss above initial ten-day VaR	69.91%	(34,955/50,000)
(d)	Loss above initial regulatory capital	8.89%	(4,445/50,000)

The stressed VaR is about 3.34 times the initial ten-day VaR and exceeds the initial capital by \$15,235.21. Under this stress scenario, the capital is exceeded more often than the 5% rate associated with a 95% VaR. As a simulation check, the exact conditional mean loss is \$70,802.45; the simulated mean is only 0.16 Monte Carlo standard errors away.

Python code, data and running instructions:
https://github.com/jesuswalking/MF-731-HW1/tree/main/HW5